

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Ch 1 — Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1-7.6)

ANSWER KEY (Version 5)

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Q1. “Biodiversity” refers to: **B — Variety of living organisms on Earth** **1 mark**

The textbook defines **biodiversity** (short for “biological diversity”) as “the variety of living organisms on Earth.” It has both ecological and economic importance for humans. Variety of habitats, number of endangered species, and an animal-only classification system are all related ideas but not the actual definition.

Q2. Which is NOT listed as an importance of biodiversity: **C — Formation of fossil fuels** **1 mark**

The textbook lists biodiversity's importance as providing **oxygen, clean water and air**, helping the **carbon cycle** and nutrient fixing, controlling pests, aiding **pollination and crop production**, and supplying **food, medicines**, and industrial materials (fibres, dyes, resins, gums, rubber, oil). Fossil fuel formation is a slow geological process, never mentioned as a biodiversity benefit.

Q3. NOT a characteristic of living organisms: **C — Crystallization and storage** **1 mark**

All living things share the characteristics of life: **movement, respiration, sensitivity, nutrition, excretion, reproduction, and growth**. The ability to be crystallized and stored is actually a *non-living* characteristic of viruses, which is one reason viruses sit on the “boundary of life.”

Q4. Scientific study of diversity + evolutionary relationships: **C — Systematics** **1 mark**

Systematics is defined as the scientific study of the diversity of organisms and their evolutionary relationships. **Taxonomy** is the narrower branch concerned only with identification, naming, and classification. Nomenclature is the naming system; morphology is the study of external structure/shape.

Q5. Correct statement about cytoplasm: **A — 90% water; site of biochemical reactions** **1 mark**

Cytoplasm is the aqueous substance found between the cell membrane and the nucleus. It is about **90% water** with many dissolved and suspended materials, and is the site for many biochemical processes; it also stores food granules/waste and is home to a variety of cell organelles.

Q6. Gain in functionality at a higher level of organization: **C — Emergent properties** **1 mark**

Emergent properties is the SLO term used in the textbook for the gain in functionalities as we move up levels of organization — e.g. from sub-cellular organelles to cells, cells to tissues, tissues to organs, organs to systems, and systems to the whole organism. Homeostasis, differentiation, and specialization are related but distinct concepts.

Q7. Benedict's test detects: **C — Reducing sugars** **1 mark**

In **Benedict's test**, Benedict's solution is added to the test solution and boiled for 2-3 minutes; a dirty green, yellow, or red precipitate confirms the presence of **glucose** (a reducing sugar). This is distinct from the iodine test (starch → blue-black) and the Biuret test (protein → pink/violet).

Q8. Protein supporting skin, bones, cartilage, tendons, ligaments: **B — Collagen** **1 mark**

Collagen is a major structural component of skin, bones, cartilage, muscles, tendons, and ligaments, providing support and strength. Keratin instead hardens hair, nails, feathers, horns, and beaks. Haemoglobin transports gases; fibrinogen is a clotting protein.

Q9. Actin and myosin, present in muscle cells: **B — Muscle contraction/relaxation** **1 mark**

Actin and myosin are proteins present in muscle cells that are involved in the **contraction and relaxation of muscles** during movement (locomotion, breathing, heartbeat, peristalsis, speech, etc.).

Q10. RNA-based enzymes are called: **C — Ribozymes** **1 mark**

Most enzymes are globular proteins made of folded amino acid chains, but a few enzymes are made of **RNA** instead — these are called **ribozymes**. Isozymes, coenzymes, and apoenzymes are all different (unrelated) enzyme concepts.

Q11. Wheat, fish, rice production & export value studied under: **B — Bio-economics** **1 mark**

The study of biology from an economic point of view is called **bio-economics**. The textbook gives production of **wheat, fish, and rice**, and studying their export value, as its examples. Biostatistics deals with data analysis, biogeography with geographical distribution, and biotechnology with manipulating organisms for benefit.

Q12. Interphase II (between Meiosis I and II): **B – Involves no further DNA replication** **1 mark**

DNA replication (S phase) occurs only once, during **Interphase I**, before Meiosis I — producing sister chromatids.

Interphase II, occurring between Meiosis I and Meiosis II, involves **no further DNA replication**; the cell simply prepares to separate the sister chromatids already present during Meiosis II.

SECTION B — Short Questions (11 × 3 = 33 Marks)

2(i) Define cytoplasm. State its water content and TWO functions. **3 marks**

Cytoplasm is the aqueous substance found between the cell membrane and the nucleus of the cell. **1 mark**

It is about **90% water**, containing many dissolved and suspended materials. **1 mark**

Two functions: it is the **site for many biochemical processes**, and it **stores food granules and waste materials** (it is also home to a variety of cell organelles). **1 mark**

OR

Why are liver cells “most metabolically active”? TWO functions. **3 marks**

Liver cells are almost round in shape, with a prominent nucleus and an abundance of cytoplasmic organelles — this makes them the most metabolically active cells of the body. **1 mark**

Two functions (any two): **1 mark each**

- Storage of glycogen, iron, and some vitamins.
- Detoxification of toxic substances.
- Production of clotting proteins of blood.
- Recycling of old red blood cells.

2(ii) List the SEVEN characteristics of all living organisms. **3 marks**

All living organisms share the following seven characteristics of life: **movement, respiration, sensitivity, nutrition, excretion, reproduction, and growth**. **3 marks for all seven correct (partial credit for fewer)**

OR

Division of labour within a cell, and among cells. **3 marks**

Within a cell, different organelles perform their own assigned roles — e.g. **mitochondria** act as the powerhouse (produce energy), **ribosomes** carry out protein synthesis, and **chloroplasts** harvest light energy to make food. This performance of a specific job by each organelle is called **division of labour**. **1 mark**

In multicellular organisms, it is not possible for billions/trillions of cells to each independently perform every life process, so cells arrange into groups to perform a given role; a group of cells performing the same function is a **tissue**. **1 mark**

Cells originating from the same zygote differentiate into unique structures suited to their role (e.g. muscle cells elongate for contraction, neurons form thin fibres to conduct messages) and **cannot exchange their functions** — extending division of labour to the whole organism. **1 mark**

2(iii) Identify the labelled structures A, B, C, D in the animal cell diagram. **3 marks**

A = Nucleus (control centre of the cell, contains hereditary DNA). **0.75 mark**

B = Mitochondrion (produces ATP; “powerhouse of the cell”). **0.75 mark**

C = Golgi apparatus (stores, modifies, and packages materials for secretion). **0.75 mark**

D = Ribosome (site of protein synthesis, studs the rough ER). **0.75 mark**

OR

Structure of the nucleolus and what happens to it during cell division. **3 marks**

The **nucleolus** is a dark-staining region inside the nucleus; it is the site where **ribosomal RNA (rRNA)** is **synthesized** and where **ribosome subunits are assembled**. **1.5 marks**

During cell division, the nucleolus **disappears during prophase** (as chromatin condenses into chromosomes and the nuclear envelope breaks down), and **reappears during telophase** once the nuclear envelope reforms around the chromatin at each pole. **1.5 marks**

2(iv) Benedict's test and Fehling's test for reducing sugars. 3 marks

Benedict's test: Take Benedict's solution in a test tube, add a few drops of the test (glucose) solution, and boil for 2–3 minutes. A **dirty green, yellow, or red precipitate** forms, confirming glucose is present. **1.5 marks**

Fehling's test: Take Fehling's solution in a test tube, add the test (glucose) solution, and heat. A **red precipitate** forms, confirming glucose is present. **1.5 marks**

OR

FIVE functions of proteins, with one example protein each. 3 marks

Any five (each protein + its function): **0.6 mark each**

- **Antibodies** — provide immunity against germs (produced by WBCs).
- **Haemoglobin** — transports oxygen and carbon dioxide in blood.
- **Fibrinogen** — blood clotting protein in plasma.
- **Actin and myosin** — contraction and relaxation of muscles.
- **Collagen** — structural support of skin, bones, cartilage, tendons, ligaments.
- **Keratin** — hardens hair, nails, feathers, horns, beaks.
- **Histone** — attaches to DNA to form chromosomes.

2(v) What are ribozymes? How do they differ from typical enzymes? 3 marks

Most enzymes are globular proteins — chains of amino acids folded into a 3D shape. A few enzymes, however, are **RNA in nature**; these are called **ribozymes**. **1.5 marks**

This means catalytic (enzyme) activity is not limited only to proteins — ribozymes show that RNA molecules can also fold into shapes capable of speeding up biochemical reactions. **1.5 marks**

OR

Why enzymes work in small quantities and can be reused. 3 marks

The structure of an enzyme may change slightly during a reaction, but it **regains its actual shape** once the reaction is complete, so it remains unchanged at the end of the reaction. **1.5 marks**

Because the enzyme is **not used up** and can be used again and again, a very small quantity of enzyme is capable of catalysing a huge amount of substrate. **1.5 marks**

2(vi) Explain 'emergent properties' with an example. 3 marks

Emergent properties refers to the gain in functionalities that occurs when moving from one level of biological organization to a higher one — e.g. sub-cellular organelles → cells → tissues → organs → organ systems → the whole organism. **1.5 marks**

Example: individual atoms combine to form molecules that can have **entirely different properties** than the atoms they contain (e.g. carbon, hydrogen, and oxygen atoms forming a sugar molecule) — in the same way, organelles combining into a cell gives rise to the properties of life (respiration, growth, reproduction) that no single organelle has on its own. **1.5 marks**

OR

Structure and function of leaf epidermis and the cuticle. 3 marks

The **upper epidermis** is a single layer of cells covered by a waxy substance called **cutin**, which forms the **cuticle**; this protects the leaf tissues and prevents excessive loss of water. **1.5 marks**

The **lower epidermis** contains tiny pores called **stomata**, each enclosed by two **guard cells** (which contain chloroplasts); the guard cells control the opening and closing of the stoma, allowing exchange of oxygen/carbon dioxide and evaporation of water vapour. **1.5 marks**

2(vii) Importance of biochemistry (any TWO points). 3 marks

Study of biochemistry is greatly helpful to explore **cell biology and anatomy**, since all structures of living organisms (cells, tissues, organs) have a specific biochemical organization. **1.5 marks**

Knowledge of biochemistry is essential to study **physiology**, because life processes such as photosynthesis, respiration, digestion, and inheritance are all explained in biochemical terms. **1.5 marks**

OR

Monomer vs polymer, using glucose and starch. 3 marks

A **monomer** is a single, simple building-block molecule — e.g. **glucose**, a monosaccharide. **1.5 marks**

A **polymer** is a large, complex molecule built from many repeating monomer units bonded together — e.g. **starch**, a polysaccharide formed from a chain of more than ten glucose (monosaccharide) units. **1.5 marks**

2(viii) A.F.A. King's 1883 observations and resulting hypothesis. 3 marks

American scientist A.F.A. King listed 20 observations in 1883 about the spread of malaria. Important observations included (any two): **2 marks**

- People who slept **outdoors** were more likely to get malaria than those who slept indoors.
- People who slept under **fine nets** were less likely to get malaria than those who did not use such nets.
- People who slept near a **smoky fire** usually did not get malaria.

Based on these observations, King suggested the hypothesis: **“Mosquitoes transmit *Plasmodium*, so are involved in the spread of malaria.” 1 mark**

OR

Qualitative vs quantitative observations, with an example of each. 3 marks

Qualitative observations are based on some **quality or characteristic** (not numeric) — e.g. observing that a leaf is “green.” **1.5 marks**

Quantitative observations are based on a **measurable value** and can be expressed in numbers, making them more accurate — e.g. measuring a leaf's length as “8 cm.” **1.5 marks**

2(ix) TWO plant and TWO animal sources of proteins. 3 marks

Plant sources (any two): beans, lentils, peas, grams, soya beans — seeds are the most common source.

1.5 marks

Animal sources (any two): milk, beef, mutton, poultry meat, eggs, fish/seafood, cheese. **1.5 marks**

OR

What is a wax? Its source and one function. 3 marks

A **wax** is a type of lipid, structurally similar to other acyl-glycerides but without the typical triglyceride form.

1 mark

Natural source: **insects** use wax to construct their **hives/combs**. **1 mark**

Function: forms the **waxy cuticle of leaves**, acting as a protective, waterproof layer that prevents water loss.

1 mark

2(x) Cell theory and THREE reasons cell division is necessary. 3 marks

According to **cell theory**, new cells originate by division of a pre-existing (parent) cell, forming daughter cells.

1 mark

THREE reasons cell division is necessary (any three): **development, growth, healing** (wound repair), and **sexual/asexual reproduction**. **2 marks**

OR

The mitotic (M) phase and its two sub-processes. 3 marks

The **mitotic (M) phase** is the part of the cell cycle during which the cell actually divides to form daughter cells.

1 mark

Its two major sub-processes are **karyokinesis** (division of the nucleus) and **cytokinesis** (division of the cytoplasm). **2 marks**

2(xi) Define biodiversity and THREE ways it is important. 3 marks

Biodiversity is defined as “the variety of living organisms on Earth.” **1 mark**

THREE importances (any three): provides **oxygen, clean water and air**; helps the **carbon cycle** and fixes nutrients; controls pests; aids **pollination and crop production**; provides **food and medicines**; provides **industrial materials** (fibres, dyes, resins, gums, rubber, oil). **2 marks**

OR

What is systematics? How does it differ from taxonomy? 3 marks

Systematics is the scientific study of the diversity of organisms and their **evolutionary relationships**.

1.5 marks

Taxonomy is the narrower branch of biology concerned specifically with the **identification, naming, and classification** of organisms — it does not by itself study evolutionary relationships the way systematics does.

1.5 marks

3(i) Draw a well-labelled diagram of the transverse section of a leaf. 5 marks

Correct diagram showing the layered structure of a leaf cross-section: **2 marks**

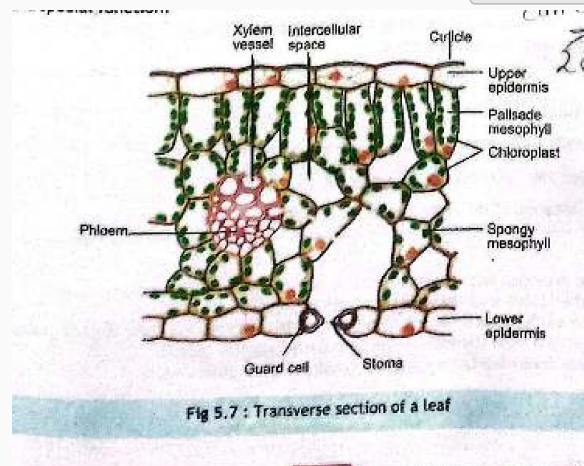


Fig. 5.7: Transverse section of a leaf (as shown in the textbook)

Any SIX correctly labelled parts (0.5 mark each, 3 marks): **cuticle, upper epidermis, palisade mesophyll, chloroplast, spongy mesophyll, intercellular space, xylem vessel, phloem, lower epidermis, guard cell, stoma.** **0.5 mark each**

OR

How is a leaf adapted for photosynthesis and transpiration? 5 marks

The **cuticle** (waxy cutin layer covering the upper epidermis) prevents excessive water loss during transpiration, while remaining transparent enough to let light through for photosynthesis. **1.5 marks**

The **palisade mesophyll** (2–3 layers of cylindrical cells packed with chloroplasts, just below the upper epidermis) is positioned to capture maximum light for photosynthesis. **1.5 marks**

The **spongy mesophyll** consists of loosely arranged, irregularly shaped cells with large intercellular air spaces, which allow efficient diffusion of CO₂ and O₂ to and from the photosynthesising cells. **1 mark**

Stomata (pores in the lower epidermis, each flanked by two guard cells) open and close to regulate the exchange of gases (CO₂ in, O₂ and water vapour out) needed for photosynthesis and transpiration. **1 mark**

3(ii) (a) Biodiversity's importance (any FOUR). (b) Systematics vs classification. 5 marks

(a) Biodiversity is defined as “the variety of living organisms on Earth.” **0.5 mark** Any FOUR importances: provides **oxygen, clean water and air**; helps the **carbon cycle** and fixes nutrients; controls pests (via insects, birds, fungi); aids **pollination and crop production**; provides **food and medicines** (mainly from plants); provides **industrial materials** such as building materials, fibres, dyes, resins, gums, adhesives, rubber, and oil. **2.5 marks (4 points)**

(b) Systematics is the scientific study of the diversity of organisms and their evolutionary relationships. **1 mark**
Classification is simply the grouping of related facts (organisms) into classes based on similarities and differences — bringing together like things and separating unlike things. Systematics goes further than classification by additionally studying evolutionary relationships among organisms. **1 mark**

OR

(a) SEVEN characteristics of life. (b) EIGHT taxonomic ranks. 5 marks

(a) All living organisms share: **movement, respiration, sensitivity, nutrition, excretion, reproduction, and growth.** **3 marks**

(b) The eight taxonomic ranks, from broadest to narrowest: **Domain > Kingdom > Phylum** (Division, for plants/fungi) **> Class > Order > Family > Genus > Species.** **2 marks**

3(iii) Detailed functions of proteins: collagen/keratin, actin/myosin, haemoglobin, antibodies. 5 marks

(a) Collagen and keratin (structural proteins): Collagen is a major component in the structure of skin, bones, cartilage, muscles, tendons, and ligaments, providing support and strength. Keratin is the component of hair, nails, feathers, horns, and beaks, making them hard and protective. **1.5 marks**

(b) Actin and myosin (muscle contraction): these proteins are present in muscle cells and are involved in the contraction and relaxation of muscles during movement — e.g. locomotion, breathing movements, blood pumping by the heart, peristaltic contraction of the gut, and speech movements. **1.5 marks**

(c) Haemoglobin (oxygen transport): red blood cells contain the red pigment haemoglobin, a protein that transports oxygen and carbon dioxide in the body via the blood. **1 mark**

(d) Antibodies (immunity): antibody proteins, produced by white blood cells (WBCs), provide immunity against germs. **1 mark**

OR

THREE biomolecule tests: iodine, Benedict's/Fehling's, Biuret. 5 marks

(a) Iodine test (starch): Add a few drops of iodine solution to the test solution or food material (e.g. a starch solution or a potato slice). A **dark blue-black colouration** confirms the presence of starch. **1.5 marks**

(b) Benedict's/Fehling's test (reducing sugars): Add Benedict's solution to the test (glucose) solution and boil for 2–3 minutes (or add Fehling's solution and heat). A **dirty green, yellow, or red precipitate** (Benedict's) or a **red precipitate** (Fehling's) confirms glucose is present. **2 marks**

(c) Biuret test (protein): Take the protein solution, add concentrated NaOH and a few drops of 1% copper sulphate solution, then shake well. A **pink or violet colouration** confirms protein is present. **1.5 marks**

3(iv) (a) Fat-soluble vitamins. (b) Structure of steroids. 5 marks

(a) The FOUR fat-soluble vitamins mentioned in the textbook are **Vitamins A, D, E, and K** — they are lipid in nature. **Vitamin D** is synthesized/derived from **cholesterol** in the body. **2 marks**

(b) Steroids are lipids that do **not** have fatty acids in their structure; instead, they have a distinctive **three-ring (fused-ring) structure**. Examples given in the textbook: **cholesterol** and **vitamin D**. **3 marks**

OR

(a) Tri-acyl glyceride vs phospholipid. (b) Why lipids yield more energy. 5 marks

(a) A **tri-acyl glyceride** is formed from one glycerol molecule bonded to **THREE fatty acid chains** (glycerol + 3 fatty acids → tri-acyl glyceride). A **phospholipid** instead has glycerol bonded to only **TWO fatty acid chains**, with a phosphate group and nitrogen base attached in place of the third fatty acid — this structural difference is what allows phospholipids to form the bilayer of the cell membrane. **2 marks**

(b) Lipids are composed of carbon, hydrogen, and oxygen, like carbohydrates, but they contain a much **higher carbon-hydrogen ratio** than carbohydrates. Because these extra C-H bonds store more chemical energy, one gram of lipid yields about **9.1 kcal** of energy — more than double the energy yielded by one gram of carbohydrate or protein. **3 marks**

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Ch 1 — Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1–7.6)

— End of Answer Key —